# Haotian Liu

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## EDUCATION

**Xiamen University (XMU), Xiamen, China** Sep. 2023 – Jun. 2027 (expected)

*B.Eng. in Software Engineering | Overall Average: 82.66/100*

**Relevant Coursework:**
Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); The Practice of Computer Programming (92).

## PUBLICATIONS

**PilotBench: A Benchmark for General Aviation Agents.**
*IJCNN 2026*, **accepted**. Co-first author. arXiv:2604.08987

**Calibrated Lexical Fusion for Noise-Robust Retrieval of Colloquial Chinese Procurement Queries.**
*ICASSP 2027*, under review. First author.

## RESEARCH PROJECTS

**CETI: Before the Agent Acts** Mar. 2026 – Present

*First Author | Research Intern at ZGC Lab | Joint collaboration with Jialong Li*

- Proposed CETI, a repository-grounded pre-coding review workflow that represents task constraints as editable **Records** linked to pinned source passages, allowing developers to revise an agent's task interpretation before code generation.
- Evaluated the workflow with 24 participants across 72 repository-task cases; the two-cycle Records condition showed the largest descriptive understanding gain (+17.2 percentage points adjusted contrast vs. FreeChat-2; HC3 95% CI [–6.0, 40.5]).
- Traced 102 confirmed request occurrences through 93 within-trajectory families, linking 48 behavioral families to patches and test evidence across 504 coding attempts on 10 tasks and 3 models.

**Long-Horizon Memory Governance for LLM Agents (LHMG)** Dec. 2025 – May 2026

*First Author | Remote Research Collaboration | School of Data Science, Fudan University*

- Developed a conflict-aware memory-governance framework for long-horizon LLM agents, combining revision graphs, No-Silent-Overwrite updates, risk-gated conditioning, and governed forgetting.
- Observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory in controlled diagnostics.

**PilotBench: General Aviation Agent Benchmark** Dec. 2025 – Mar. 2026

*Core Contributor | Remote Research Collaboration | CESS, Beihang University*

- Co-developed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases; evaluated 41 models with Pilot-Score, a composite of regression accuracy and instruction following and constraint compliance.
- Identified a precision–controllability dichotomy: traditional forecasters reached MAE 7.01, while LLMs achieved 86–89% instruction-following at 11–14 MAE; performance degraded in high-workload flight phases.

**NeuroMark: Multimodal Reliability Evaluation for Neuropsychiatry** Sep. 2025 – May 2026

*Research Contributor | Remote Research Collaboration | School of Data Science, Fudan University*

- Contributed to a multimodal neuropsychiatry benchmark that jointly evaluates text-based clinical reasoning and brain-imaging interpretation across 2,379 clinical questions and six imaging tasks.
- Evaluated 20 LLMs and 16 VLMs; helped develop SCR, EA, DDI, and JTI metrics to measure constraint compliance, effective accuracy, difficulty sensitivity, and cross-modal reliability.

## INDUSTRY EXPERIENCE

**ProcuraClaw: Procurement Search System** Nov. 2025 – Apr. 2026

*First Author | H World Group*

- Built a deployed B2B procurement-search system for a global hotel group; designed a four-path hybrid retrieval pipeline (exact, fuzzy, synonym, and selective semantic via BGE-M3) fused with RRF, and a hotel-profile-driven quality function across five operational dimensions.
- Implemented online personalisation with Thompson Sampling and Posterior Dropout; achieved MRR 0.95, NDCG@5 0.91 (175 queries) with 6.2%/9.8% first-hit CTR gains in A/B tests; work written up as a paper under review at ICASSP 2027.

## RESEARCH AND TECHNICAL SKILLS

**AI and Programming:** LLM agents and memory, multimodal learning, AI for software engineering and HCI, RAG and information retrieval, benchmark and evaluation design; Python, PyTorch, Hugging Face, Spring Boot, SQL, Java/Java EE, C/C++, Unity.
**Research Methods and Tools:** Controlled experiments and human studies, ablation and error analysis, academic writing and presentations; Git, Linux, Docker, and LaTeX.
**Languages and Tests:** Chinese (native); English (TOEFL iBT 90); Japanese (reading knowledge).